

Refractive Gravity: An Ontological Picture and the Foundations of Mathematical Formalism

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2026

Zenodo DOI: 10.5281/zenodo.21325652

Abstract

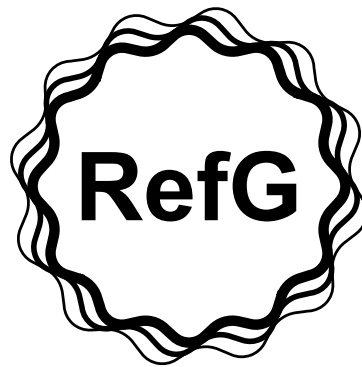
This paper presents an ontological account of Refractive Gravity (RefG) that integrates several lines of scientific work and inquiry into a single physical picture. RefG is a research program under stepwise development. The central aim of the text is to offer the reader an intuitive picture of the fundamental organization of the Universe; the formalism and completed calculations are available in the public scientific archive on Zenodo.

The current status of the ideas and results discussed in this paper is given in the final section, “Current Status of the Research Program.”

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Introduction

Modern physics resembles an intricate puzzle. Its individual pieces have been calculated with extraordinary precision and tested experimentally, yet they still resist assembly into a single physical picture. The Standard Model of particle physics and the geometrical description of gravity do not fit fully within one language. In the interiors of black holes, classical geometry reaches singularities; standard cosmology explains galactic rotation curves and the regularities of modified Newtonian dynamics (MOND) through dark-matter halos, and the accelerated expansion of the Universe through dark energy.

Quantum randomness, the nature of time, the hierarchy of particle masses, the fundamental constants, and the relative weakness of gravity belong to the same larger picture. Each has been studied in depth, but their common physical basis remains unseen.

The history of physics offers many attempts to bring these problems together. Albert Einstein sought the answer in the geometry of spacetime, while John Archibald Wheeler connected geometry and information through the principle of *It from Bit*. Andrei Sakharov described gravity as an effect induced by quantum fluctuations of the vacuum; Ted Jacobson derived it from the thermodynamics of horizons; and Erik Verlinde interpreted it as an entropic and holographic phenomenon. Roger Penrose treats spacetime, cosmology, and quantum physics as a single geometrical problem.

Their mechanisms differ, but the central question is shared: can many phenomena be described from one foundation?

This paper assembles ideas developed over many years into a unified logical structure and presents the ontological framework of **Refractive Gravity (RefG)**.

Modern physics generally begins with coordinates, time, and fields already in place. RefG steps back and asks: before speaking of space, time, or mass, what creates a stable and measurable distinction in nature?

In 1905, Einstein abandoned absolute time and defined simultaneity operationally through clocks and light signals (Einstein, 1905). RefG poses that operational question at an earlier level: what physical condition creates the clock and the signal on which those operations themselves depend?

The argument rests on three fundamental principles:

- **First:** Nature has only one ontological foundation—one primary identity. Spatial location, clock time, the measurable Universe, and the physically manifested medium arise through its resonant self-distinction. Fields, forces, and particles are phase, pressure, shear, rotational, or topological modes of that manifested medium.
- **Second:** The stable unit of matter is the *oscillon*—a locally confined, self-sustaining resonant form of the manifested medium. Its intense internal dynamics leave a persistent pressure-deficit imprint in the surrounding medium.
- **Third:** Gravity is the refractive response of the manifested medium. The oscillon’s oscillatory action propagates through the medium, lowers its static pressure, and creates a long-range deficit imprint. The stable profile of this mechanism, phase-locked to its source, is read as ordinary gravity; the coherent component released during a rapid reconfiguration of the source is observed as a gravitational wave. Attraction is motion within an effective geometry altered by an oscillation-generated pressure deficit.

The Michelson–Morley experiment of 1887 found no ether wind of the kind expected from the Earth’s motion and placed stringent constraints on a stationary ether independent of matter (Michelson and Morley, 1887). RefG proposes a different picture: space, time, the measuring instrument, and the observer are all modes of a medium generated from one foundation.

Under uniform motion, form is transmitted with wave and phase symmetry. On the Lorentz-covariant branch, this process leaves no trace of energy loss. Lord Kelvin’s intuition is extended here: matter is itself a form of the medium, while the measurable Universe takes shape through the self-distinction of the primary foundation.

This text brings the calculations collected in the public Zenodo archive into an intuitive physical picture (Lotishvili, 2026a,b). It connects the self-distinction of the foundation with measurability, the pressure deficit of mass with cosmological horizons, and the origin of particles with the topology of quantum numbers. Earlier programs also pursued medium-based unification, including Winterberg’s Planck aether (Winterberg, 2002). RefG places its central emphasis on the physical origin of measurability and the common resonant locking of the particle population. Its principal contributions are as follows:

1. **Population-locked time:** Self-distinction first produces a pre-clock ordering of events; mutual locking across the particle population then turns that order into a local clock rate. Each mode follows the shared rhythm through its own stable ratio. Time dilation is a collective reconfiguration of this local network. This population-based definition of temporal rate is a distinctive element of RefG.
2. **Two-channel accounting:** Gravitational mass is, by definition, a readout channel through the medium. The oscillon’s internal state is retained as a physically separate quantity, and RefG distinguishes the two axiomatically.
3. **Separation of size-dependent and local properties:** On the selected first-order biconformal branch, size, externally read mass, and clock rate follow the pressure change directly, whereas the coordinate propagation of light follows it quadratically. RefG relates this distinction to the different roles of nodes and the connections between them.
4. **The Koide three-phase structure:** In the three-phase map of the charged leptons, the second branch of the ninth-order return cycle fixes the phase at two ninths of a radian, while equalization of the internal channels fixes the modulation depth at the square root of two.
5. **A node-based interpretation of the fine-structure constant:** In the RefG picture, the fine-structure constant is a dimensionless, invariant ratio between the electric and magnetic responses of the node network of the manifested medium. It describes the relative strength of the smallest electric flux and the transverse oscillation of the photon supported by the same network.
6. **The scope of process time:** Coordinate time and process time are clearly distinguished. Process time serves as an explanatory language for physical phenomena, while cosmological observations are calculated in coordinate time.
7. **Two readouts of one operator:** RefG’s unified spectral map associates short-wavelength modes with the particle readout and long-wavelength modes with the cosmic-web readout.

The novelty of each link in this chain must be assessed separately. Medium-based unification programs have precedents, and a similar separation between linear and quadratic dependences also appears in polarizable-vacuum models (Dicke, 1957; Puthoff, 2002). RefG derives these two powers from the distinct roles of nodes and connections and joins them to two-channel mass accounting.

The three-phase parametrization of the Koide relation and its geometrical reading continue the work of Brannen and Foot, while Shulga developed a different holonomy-based derivation of the phase corresponding to two ninths of a radian (Brannen, 2010; Foot, 1994; Shulga, 2026). RefG connects this structure to phase indistinguishability, the ninth-order return cycle, and equalization of the internal channels.

The relaxation-based reading of dark energy and the knot-based argument for three-dimensionality likewise draw on established lines of thought (Klinkhamer and Volovik, 2008; Whitrow, 1955; Berera et al., 2017). The distinctive backbone of this work is their connection to the birth of measurability, a population-wide shared temporal rate, two-channel accounting, and the two scale-dependent readouts of a single spectral operator.

In this picture, dark energy is associated with the remnant of the historical relaxation of the self-distinguished cosmological background, while three-dimensionality is associated with knot closure and topological stability.

Modern information, symbolic, and software tools support the systematic computational testing of the intuitive ideas developed in this research.

The text invites the reader to follow this logical chain and see how a single physical picture connects phenomena that otherwise appear separate. The scientific value of the framework will be determined by its further formalization and observational testing.

1 Ontological Foundation

1.1 One Foundation and Its Manifested Medium: Substance, Emptiness, and the Dynamics of Existence

Modern fundamental physics describes the universe through several independent fields. In the Standard Model, each field has its own dynamics and rules of interaction. RefG begins instead from a single principle: there is one ontological foundation—one primary identity—whose resonant self-distinction gives rise to a multimodal physical medium. Fields and forces are phase, pressure, shear, rotational, and topological modes of this common medium.

The idea of a unified foundation has a long history in physics. William Clifford pictured matter as small, moving *hills* of spatial curvature (Clifford, 1876). Clifford’s picture was purely geometric; the medium of RefG carries pressure, shear, and resonant channels together with its geometric response.

A physical example of distinct particle modes emerging from one carrier is provided by Levin and Wen’s *string-net* condensation model. Within the same bosonic system, collective excitations arise as gauge bosons and fermions and, in suitable phases, behave like photons, electrons, gluons, and quarks (Levin and Wen, 2005). Their model begins with an already structured quantum system. RefG associates the same possibility with the self-distinction of a single pre-spatial foundation.

The simplest dynamical picture of the manifested medium lies between two limiting states. At one limit is maximal substance, where pressure, mass–energy potential, and operational density are extremely high. At the other is the zero limit, where interaction and measurability disappear. Physical reality is the bounded resonant dynamics between these limits. *Nodes of existence* arise within this dynamics.

One foundation unfolds into many properties in the manifested medium. The medium can be compressed, twisted, set into vortical flow, and made to carry a wave. Water makes the idea intuitive: one fluid simultaneously supports pressure, flow, and surface tension, although these properties are not interchangeable. In the same way, the RefG medium combines phase, pressure, shear, and topological responses. This multichannel structure underlies the formation of electric charge, spin, and the particle spectrum.

The manifested medium has a quiescent background state, denoted formally by $\varphi = 0$. Every measurable location is defined by relations among nodes of the self-distinguished medium. The local excitation level, the operational density of nodes, and the strength of their connections may all vary. In this language, a pressure drop, or rarefaction, is an increase in the interval between nodes that cannot be measured directly.

Here *pressure* means the effective pressure–energy state of the medium. Pressure rises above

the background, while below it a deficit branch appears. Gravity in this paper is read through that deficit profile. Classical fluid pressure is its intuitive analogue. From outside, a change in the deficit appears as a change in clock rate, signal propagation, operational size, and effective mass.

1.2 From One Existence to Multiplicity: The Origin of Measurability and Space

A completely undifferentiated foundation is one indivisible primary identity. Its resonant self-distinction establishes the first distinguishable relation; at that same threshold, location, clock-measurable duration, quantity, and counting arise.

In the undifferentiated state, this identity retains its full unity. Place, size, mass, clock duration, beginning, and end are properties of self-distinguished processes and arise with them.

Measurable physics begins with resonant self-distinction. At the first stage, resonance is a self-consistent, closed relation among states of the foundation. This pre-clock structure establishes an ordering of events; comparison among stable cycles then gives rise to clock time.

Before self-distinction, this origin is called the **primary identity** or the **foundation**. A persistent difference generated through self-distinction takes the form of a node, a phase displacement, or a pressure deficit. A **node** is a distinguishable state of the one foundation; an **imprint** is the change left by that state; and a **relation** is the comparison of one state with another. A multiplicity of nodes consists of mutually distinguishable manifestations of one identity, held together as a whole by a common resonant order.

Quantity begins at this threshold. A difference that can be read persistently produces countable physical units. Manifestations become distinguishable from one another, retain their own imprints within the common order, and can therefore be counted.

The primary unit of measurement is not an isolated object but a distinguishable relation. A completely undifferentiated identity contains no second term of comparison. Resonant self-distinction produces two mutually distinguishable states of the same foundation and the first connection between them; counting begins at precisely this point. The foundation does not break into separate pieces. It retains its unity, while its persistent, mutually distinguishable manifestations are read as multiplicity.

In informational language, the first preserved difference is analogous to a bit. A distinguishable manifestation is read as 1, while the absence of its imprint in a given relation is read as 0. Different arrangements of these signs gradually form more complex structures, just as mutually distinguishable states of the foundation generate the diversity of the measurable universe.

The relation between two distinguishable nodes is the physical seed of space. Delay in their connection is the seed of time; their extension across the network gives rise to operational size; and the energetic tension required to preserve the difference provides the internal basis of mass. External gravitational mass is read from the deficit profile that this tension creates in the medium.

Thus the resonant self-distinction of one primary identity produces distinguishable states. Their preserved differences create quantity, while their mutual relations create space. The relational ordering of changes establishes a pre-clock sequence, and comparison among stable cycles produces clock time. Extension through the network determines size; stable energetic confinement and its deficit imprint determine internal mass and mass as read from outside.

This view also clarifies the nature of the observer. We and our measuring instruments are complex structures built from the same nodal network. The idea echoes Bridgman's operationalism, according to which a physical concept is defined by the operations through which it is measured (Bridgman, 1927). RefG asks the deeper question: what physical condition makes measurement itself possible?

In this picture, the universe is a continuous physical process generated by the self-distinction of one foundation. The foundation has no separate spatial location, because place itself arises only

from relations among its mutually distinguishable manifestations. Every measurable location is a different manifestation of this one identity.

The observer and the measuring apparatus belong to the same medium, so the interval between nodes is read through its own physical imprint. A distinguishable relation between two nodes produces space. Below that threshold, the difference remains an internal state of the medium; beyond it, the difference appears as a change in time, size, mass, and signal propagation.

RefG posits a minimum threshold of operational distinguishability, denoted by ℓ_* . Its physical content is the formation of a relation preserved as an independent, stable imprint. Below this threshold, spatial separation does not exist as a measurable property. At the threshold, the first operational interval is born; above it, that interval is read through the signatures of pressure, clock rate, and signal propagation. The Planck length is a natural physical scale, while ℓ_* is the hypothetical threshold for the formation of a stable record. The microdynamics of the medium must determine whether the two coincide.

The operational basis of the measurable universe is therefore the first stable relation that makes multiplicity possible. The foundation is one; the universe is the network of its mutually distinguishable nodes and connections.

The same object occupying different locations at different times provides a simple image of this idea. If its entire path is shown in a single picture with the time labels removed, the result is an extended trace: the identity is one, while the marked locations are many. RefG takes this picture to a deeper level. Persistent self-distinctions of the foundation create the very locations that can then be counted and compared.

A related idea appears in Mohrhoff's quantum ontology, where spatial extension and particle multiplicity arise from one non-spatial and atemporal origin. In that picture, space is constituted by reflexive relations, and fundamental particles are read as numerically identical with the same underlying entity (Mohrhoff, 2014). RefG joins this motif to the hypothesis of resonant self-distinction. The transition must generate a medium, clock time, and mutually distinguishable modes whose energy, momentum, and charge are additive.

The atemporal status of the foundation and the age of the measurable universe belong to different conceptual levels. Age belongs to the self-distinguished regime; the completely undifferentiated foundation-state is atemporal.

Mutual locking among stable oscillons turns the pre-clock sequence into a clock-measurable rate.

At this boundary, the primary physical content of gravity also appears. When the connection between nodes weakens, it is read from outside as a fall in pressure, a slowing of clocks, and a change in signal propagation. At macroscopic scales, this coordinated change is called a gravitational field.

1.3 Nodal Transmission: Waves, Time, and Self-Calibration

Self-distinction of the foundation gives rise to a unified network of nodes of existence. Each node is a functionally distinct state of the same foundation: it preserves an imprint of change and transmits it through the network. The operational density of nodes describes how closely these manifestations can be resolved. At the first level, adjacency is direct physical coupling; stable order among such couplings gives rise to spatial proximity and operational distance.

A wave is the sequential transmission of state and phase through this network. Light is a freely propagating transverse phase imprint, whereas an oscillon is a locally confined, self-sustaining pattern of the same transmission. As matter moves, this confined form is relayed from node to node and re-establishes the same phase coherence at each new location. Light and matter are therefore propagating and confined modes of one medium.

Every transmission establishes a relational sequence of events. Repeated phase relations organize this sequence into a pre-clock order, while comparison among stable oscillon cycles turns it into clock time. The local rate of time is the collective rhythm of the entire network: it

records both the rate at which the medium transmits state and the rate at which an oscillon completes its cycle.

Thermal insulation in double glazing offers a clear analogy. Evacuating the gas between the panes suppresses convection and, at sufficiently low pressure, also reduces molecular thermal conduction, although thermal radiation still propagates. The essential point is that less heat is transferred per unit time. RefG extends this intuition to the nodal medium and relates the rate of transmission to its density and coupling strength.

Imagine an observer whose body, clock, and ruler are themselves made from this flow of heat. Such an observer could not isolate the carrier gas using a standard of measurement independent of the observer’s own physical mode; the gas would be inferred only from changes in pressure and heat transfer. In RefG, the observer, the measuring apparatus, light, and matter are all modes of one manifested medium. We therefore read the state of the foundation only through its collective pressure imprint—through changes in clock rate, signal propagation, operational size, and effective mass.

In a deficit region, the operational density of nodes falls, pressure decreases, and the common rate of nodal transmission slows. A distant observer reads this single change as slower wave propagation, longer oscillon cycles, reduced operational size, and an altered effective-mass readout. The local observer’s clock, ruler, body, and light signal are reconfigured with the same medium; in local units, the observer continues to measure the usual rate, the usual size, and the same invariant local speed of light.

The same confined resonant pattern preserves its topological identity and internal phase relations in both dense and rarefied nodal configurations. What changes is its realization in the medium—how broadly it extends across the operational network and what mass is read from a distance. The local ruler belongs to the same configuration and therefore cannot register a common change of scale relative to itself. Comparing processes that have traversed different environments, however, reveals the difference as a real physical change in time, size, and mass.

The operational density of nodes, coupling strength, and transmission rate are interdependent properties of one medium. In a deficit region they vary coherently and together define a unified pressure-mediated dynamics. On the physically admissible low-energy branch, transmission must be isotropic, lossless, and Lorentz symmetric: the local speed of light remains invariant, and uniform motion leaves no trace of drag.

1.4 The Shared Local Rhythm and Resonant Locking

The shared rhythm of local time emerges through mutual locking among stable oscillons—the wave forms of matter. Within a unified resonant network, their internal frequencies settle into fixed, stable ratios, while the internal coordination of the local network establishes the rate of time.

No physical frequency can be assigned to the undifferentiated foundation; a vibrational spectrum forms only in the self-distinguished medium. This three-dimensional oscillatory medium sets the boundaries and determines which modes can exist. The sound of a bell likewise arises from modes resonating together across the whole body. In the same way, the frequencies of elementary particles are established through mutual balance in a common medium. The medium’s microdynamics selects the admissible modes, and their mutual coordination produces a stable, locked configuration.

Every particle mode has its own frequency, yet all follow the same local rhythm and preserve stable ratios to one another. These ratios need not correspond to integer harmonics; the resulting spectrum instead resembles that of a three-dimensional, mildly anharmonic resonator. Each mode retains its identity, while a rhythm incompatible with the network loses phase coherence and cannot persist as a stable particle form.

In a low-pressure region, the shared rate of the entire local resonant network changes. A dense nodal configuration corresponds to strong effective coupling and a faster common rhythm;

in a rarefied medium, the collective transmission rate falls. The operational density of nodes, coupling strength, and clock rate are interrelated properties of one medium, whose joint effect is expressed by its pressure state.

All local clocks and particle processes follow a change in the common rate together. A local observer continues to read time as usual, while the network’s internal dimensionless ratios—including the fine-structure constant and particle-mass ratios—remain invariant.

The classical precursor of this relational view is Ernst Mach’s principle (Mach, 1883). RefG gives it a local form: the common rhythm is set directly by the local resonant network, and physical action therefore remains local.

In the RefG map, the fine-structure constant is the dimensionless, invariant ratio between the electric and magnetic responses of the nodal network.

As a wave passes from one pressure regime into another, it adjusts to the new medium. Its observed frequency is determined by the effective metric. In RefG language, comparison between the local rhythms of source and receiver is a translation of phase; cosmological redshift is read from the ratio between the scales at emission and reception.

A wave does not carry a frozen, absolute frequency along its path. In each region, its phase adjusts to the common local rhythm. A compatible imprint locks into the new resonant network; an incompatible one loses phase coherence or scatters, and its energy passes into other compatible channels of the medium.

The rate of time is therefore the physical state of the local resonant network. When pressure changes, the entire network reconfigures collectively: all compatible frequencies change together, while their dimensionless ratios remain invariant. The validity of the common reconfiguration is tested by this invariance.

1.5 The Oscillon, the Pressure Deficit, and the Origin of Mass

In this framework, electrons, protons, and other particles are treated as oscillons. An oscillon is a stable wave structure confined within the medium—a highly organized form that sustains itself. The historical precursor of this idea is Lord Kelvin’s vortex-atom model (Thomson, 1867). That particular program belongs to history, but its central intuition remains: a particle may be a stable, dynamical form of the medium itself. RefG develops this intuition in modern resonant and topological language.

The confined form of an oscillon is local, but its oscillatory action does not stop at its local boundary. Each cycle transmits a phase and pressure change to neighboring nodes; the imprint propagates from node to node and spreads throughout the connected, boundaryless medium. As it covers an ever larger region, its amplitude decreases with distance.

When an oscillon oscillates or rotates intensely, or confines energy within its form, the dynamical activity of the medium increases and its static pressure falls. Bernoulli’s picture makes the direction intuitive (Bernoulli, 1738): greater dynamical intensity is accompanied by lower static pressure. On the selected weak branch of RefG, this analogy corresponds to a pressure–energy balance identity. During the formation or reconfiguration of a source, RefG relates the broad redistribution of oscillatory energy to relaxation of the cosmological background; Chapter III develops the thermodynamic content of this connection.

In RefG, ordinary gravity and a gravitational wave are two regimes of the same oscillatory mechanism of the medium. Ordinary gravity is a persistent wave profile phase-locked to its source: the oscillation extends across the entire connected network, but once the profile is stationary it no longer produces a net outward flux of energy. A gravitational wave is the coherent oscillatory change released from that profile when the source reconfigures rapidly, carrying energy outward. An ordinary wave and a tsunami share the same ocean and propagation mechanism; they differ in regime, scale, coherence, and transported energy. Observations of gravitational waves establish that a propagating disturbance periodically changes the separation between freely falling test

bodies. RefG reads this large traveling wave as the outgoing manifestation of the same channel whose stationary profile constitutes ordinary gravity.

Through its active existence, an oscillon creates a pressure-deficit profile in the manifested medium. In this map, the deficit is the direct external profile of gravitational mass: its integral readout yields the system's gravitational charge. The more intense the oscillon's internal dynamics, the deeper the deficit and the stronger the refractive gradient. The oscillon's internal reserve is accounted for separately from this external readout.

The local depth of the deficit and the total mass read from a great distance are not the same quantity. The first describes the central gradient and the state of compactness; the second combines the active core, the dressing supplied by the medium, and the negative contribution from binding. After a merger, the local deficit may therefore deepen even as the total mass read from afar decreases by the energy carried away in radiation.

This view has a well-known parallel in the history of physics: John Archibald Wheeler's program of *mass without mass*, in which a geon is a massive configuration held together by the energy of its own field (Wheeler, 1955). Wheeler's geons gradually radiate their energy away; RefG associates oscillon stability with resonant locking and topological protection.

The term *effective refractive mass* means the mass of one oscillon as read from outside in a given environment. Its internal state is determined by its topological class, phase relations, and intrinsic energy. When the background changes slowly, this internal structure preserves its identity, while the pressure of the medium changes the oscillon's operational size and externally read mass. The total mass of the complete system as read from afar is determined by the full deficit profile. Externally read mass is therefore determined by the medium's refractive response.

At this fundamental level, gravity is the refractive deflection of the wave form of one oscillon within the pressure deficit created by another. A slack string provides an intuitive image of a gravitational well: the locally reduced pressure-energy state of the manifested medium weakens the conditions of propagation. The deficit profile shapes the effective geometry of the medium, and the particle's wave form moves refractively through that geometry.

An oscillon creates a pressure deficit, and the deficit in turn changes its effective mass, operational size, and coupling to the medium. RefG reads this two-way feedback as a mechanism of dynamical equilibrium: the material form changes the medium, and the altered medium determines the conditions of the form's external physical manifestation.

2 Gravity as Refraction

2.1 Three Concurrent Operational Changes

The pressure state of the manifested medium falls locally around an oscillon. In the nodal description, this corresponds to a lower density of nodes and weaker physical coupling. An external macroscopic observer reads this unified physical state through three coordinated changes. **First: a reduction in the coordinate speed of light.** Conditions of collective transmission change within a pressure-deficit region, so light and other wave signals propagate more slowly in the coordinate description. The locally measured speed of light remains invariant; what changes is the rate read in the coordinate language of a distant observer.

Second: a reduction in the operational size of the oscillon. The oscillon preserves its fundamental topological form, while a rarefied medium entails reduced active participation within the nodal configuration. In the external description, this change appears as a reduced operational size and an altered effective-mass readout.

Third: a reduction in the rate of the local resonant clock. A clock is a real system built from oscillons, and its cadence depends on local nodal connections. In a rarefied medium, this coupling weakens and the clock slows relative to the coordinate time of a distant observer. The local clock continues to count its own cadence as usual.

These three changes are three operational readouts of a reduced-pressure state. They do not all depend on the change in the medium in the same quantitative way.

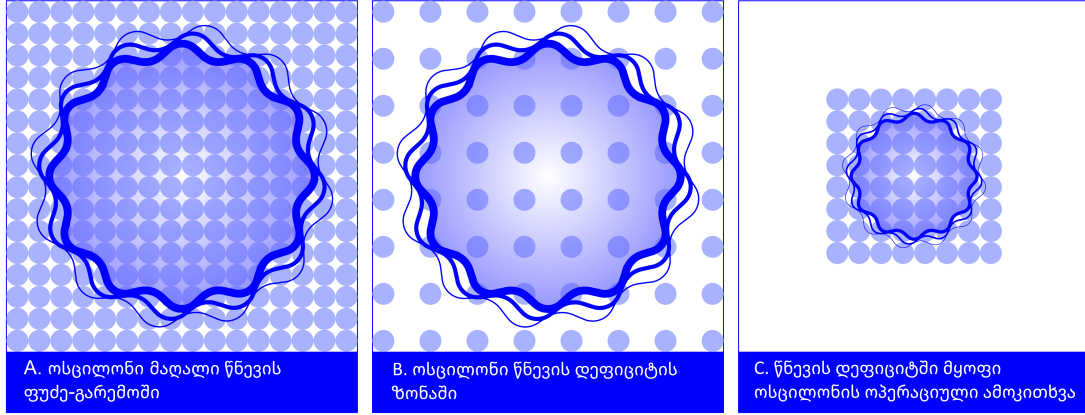


Figure 1: Operational readout of an oscillon in a pressure-deficit region.

- (A) In a high-pressure manifested medium, the operational density of nodes is high; physical coupling is strong, and the oscillon’s local processes run rapidly.
- (B) In a pressure-deficit region, the operational density of nodes falls; the operational separation between them increases, the transmission rate decreases, and local time slows.
- (C) In the external readout, an oscillon within the pressure deficit is read as having a slower clock rate, a smaller operational size, and an altered effective mass; the enlarged internal interval does not enter directly as a measurable quantity.

Separation among nodes does not create a *second space*. The interval between nodes is not an independent physical object; within this framework, what ordinary language calls “empty space” is read through the pressure variable. The operational manifestation of a rarefied medium is a local reduction in its pressure state.

2.2 The Quadratic Relation and the Factor of Two

The three operational changes do not develop with equal strength. On the selected weak-field branch of RefG, operational size, the effective mass of an individual oscillon, and the rate of a local clock follow the change in pressure directly. In the working map of the same branch, the coordinate propagation of light follows the joint change in node density and coupling. The response is therefore quadratic, and a factor of two appears.

Similar powers also occur in polarizable-vacuum models (Dicke, 1957; Puthoff, 2002). RefG relates them to the ontological distinction between nodes and connections.

As the deficit deepens, the operational size and externally read effective mass of an individual oscillon decrease. The total gravitational mass of the system generating the deficit, by contrast, is the integral readout of the entire profile at large distance. Metric filters on the selected biconformal branch connect these two scales.

Four quantities must be kept distinct:

1. the oscillon’s internal state and intrinsic energy;
2. the externally read effective mass of an individual oscillon in a given background;
3. the profile of the deficit source created in the manifested medium;
4. the total gravitational mass of the complete system as read from a great distance.

These four levels form two balances. One part of the internal reserve is visible in the external readout, while another remains within strong confinement. The external readout combines the active core with the deficit layer formed around it—the medium-borne dressing. At large distances, both appear as one gravitational mass. The local energy of a test body and the mass of the complete system are therefore quantities read from the same configuration at different scales.

A colored, ice-like crystal immersed in clear water offers a vivid image of this self-consistent dynamical process. The crystal is a locally bound form of the same water, just as an oscillon is a local resonance of the common manifested medium. As the configuration forms, transforms, and relaxes, part of its energy passes from local confinement into a state distributed broadly through the medium—as the color of a dissolving crystal spreads through the water. Energy is redistributed among different channels of the system.

When the shape of a source changes rapidly—during nonuniform acceleration, merger, or relaxation—many oscillatory contributions combine coherently and generate a strong outgoing wave. On the tensor branch, its leading observable component is tied to a quadrupolar change: an asymmetric reconfiguration of the source’s shape and mass distribution. Once the surge has passed, a new stationary gravitational profile is established.

When two compact gravitational sources merge, their separate oscillatory mechanisms lock into a single configuration. Their combined dynamical activity lowers the local pressure of the manifested medium more strongly, deepens the central deficit profile, and increases the negative contribution from binding. The rapid reconfiguration emits a strong gravitational wave, while the remnant’s total mass as read from afar decreases by an amount corresponding to the energy carried outward. A deepening local deficit and a reduction in the total mass inferred at large distance are therefore different readouts. The remnant mass and radiated energy are connected by one energy balance, whose exact form must be determined by a complete dynamical solution.

During relaxation, the strong outgoing wave spreads into an ever broader region of the medium, and the process approaches a new limiting state. Local measuring instruments are reconfigured with the same medium.

On the selected biconformal branch, temporal and spatial scales change in opposite directions as the deficit deepens. In the weak-field expansion of this branch, the sign and magnitude of the potential agree with the Newtonian potential, while the standard result of general relativity is recovered through first post-Newtonian order: the parametrized post-Newtonian coefficients satisfy $\beta = \gamma = 1$.

The duration of a complete clock cycle is read as the ratio of operational size to coordinate speed. If size is reduced by a factor of two while coordinate speed falls by a factor of four, the duration doubles—the clock runs at half its original rate.

On the same branch, the first post-Newtonian results agree with the standard values for light deflection, Shapiro delay, gravitational redshift, and perihelion precession (Bertotti et al., 2003; Will, 2014). The Cassini radio-link experiment constrains departures of the spatial coefficient from its general-relativistic value to roughly two parts in one hundred thousand. Historically, the factor of two contributed by the spatial component completes Einstein’s 1911 half-result to the full general-relativistic deflection (Einstein, 1916; Shapiro, 1964). Higher-order Solar System effects are determined by the complete field equations.

2.3 Refractive Attraction, Inertia, and the Absence of Friction

An oscillon, as a wave structure, exists within gradients of pressure and coordinate propagation speed in the manifested medium. Its phase propagates more rapidly on the denser side and more slowly on the rarefied side, so its wavefront bends refractively toward the region of lower pressure. In the weak, static, large-scale limit, the deficit created by an object shapes an effective geometry, while wave propagation through that geometry appears as gravitational motion.

In RefG, the fall of an apple is pictured as follows: the wave forms of its atoms bend toward the region of lower pressure and slower coordinate propagation created by the Earth. Familiar attraction is the refractive response of the medium.

In formal language, the gravitational source is the pressure deficit projected into the manifested medium. It is the component of the pressure–energy balance that an external observer reads as attraction. Microscopically, it is the stress profile imprinted on the medium by the oscillon’s internal dynamics.

A gravitational well is a three-dimensional region of reduced pressure in the manifested medium. The mass of a bound system is lower than the sum of its freely separated constituents by the mass equivalent of the binding energy.

GW150914 provides an observational example: gravitational waves carried away energy equivalent to roughly three solar masses (Abbott et al., 2016). In general relativity, this is the standard balance of binding energy. RefG describes the same event through one pressure mechanism. The merger of two oscillatory sources lowers the local pressure more strongly and deepens the remnant’s central deficit profile, while the mass–energy difference between the initial system and the remnant, as read from afar, is carried through the medium by a strong coherent oscillation. Its leading observed component is a transverse–traceless (TT) wave. GW150914 supplies an essential observational basis for this picture; testing the distinguishing relation requires a complete dynamical calculation that derives the remnant’s total mass at large distance, the energy of the outgoing flux, and the waveform from one initial state.

On the static compact branch, the projected deficit source is read as an effective violation of the null energy condition (NEC). In the RefG map, this stress belongs to the confinement channel of the manifested medium.

RefG associates inertia with deformation of the deficit imprint during acceleration. The pressure profile ahead of and behind the oscillon becomes asymmetric, and this asymmetry creates a reaction opposed to the acceleration. Under braking, the picture reverses as in a mirror.

For a finite-energy Lorentz-covariant oscillon, when the appropriate boundary balance is satisfied, inertia is determined by the total energy of the system. This follows from the results of Noether and Laue (Noether, 1918; von Laue, 1911). Under the same conditions, inertial and gravitational masses agree at leading order. The MICROSCOPE experiment tests this universality at the level of 10^{-15} (Touboul et al., 2022).

On the branch of a uniformly moving Lorentz-covariant solution, the wave pattern is transmitted symmetrically and leaves no dissipative imprint. RefG associates frictionless motion with this regime of phase transmission.

Motion is a wave-like relay of form. Its limiting speed is set by the maximum rate of phase transmission between nodes of the manifested medium. On the Lorentz-covariant branch, this limit coincides with the null cone of the effective metric. The finite speed of light arises from the same operational network.

2.4 Local Unobservability and the Principle of Relativity

A local observer in a gravitational field reads time and size as usual because the observer’s clock, ruler, and body arise from the same foundation. As the medium changes, the entire local system reconfigures together. If that change is much slower than the internal processes, it is adiabatic: the mode is preserved, and no sudden transition to another state occurs. Dimensionless ratios, including the fine-structure constant, remain invariant under this common reconfiguration.

This picture is subject to firm experimental bounds. Gravitational redshift has been tested across many regimes, from the Pound–Rebka experiment to Gravity Probe A and the Galileo tests (Pound and Rebka, 1960; Vessot et al., 1980; Delva et al., 2018; Herrmann et al., 2018). Quasar spectra and atomic clocks independently constrain the stability of dimensionless ratios; in cosmological comparisons, the typical bound on $|\Delta\alpha/\alpha|$ is of order 10^{-6} . RefG reads these facts as physical manifestations of local self-calibration.

The null result of the Michelson–Morley experiment corresponds to common self-calibration, whereas the LIGO signal is a tensor response that differs by direction. If the medium changes uniformly throughout a region, the ruler, clock, and propagation of light all reconfigure together; a local observer sees no difference. Modern tests constrain directional differences in the speed of light to the order of 10^{-17} . A gravitational wave, by contrast, changes different axes with opposite signs and therefore leaves a measurable imprint in a detector.

FitzGerald–Lorentz length contraction is a historical precursor of this logic (FitzGerald, 1889; Lorentz, 1892). In RefG, the compensation arises because measuring instruments are themselves wave forms of the same medium. In the local Minkowski limit, the tensor wave propagates at the speed of light and produces a transverse–traceless (TT) detector response.

Comparison of the gravitational and electromagnetic signals from GW170817 constrains the difference between their speeds to the level of 10^{-15} (Abbott et al., 2017). The orbital decay of binary pulsars tests the energy loss from quadrupolar radiation to an accuracy between 10^{-4} and 10^{-3} (Weisberg and Huang, 2016; Kramer et al., 2021). A possible scalar wave would add a change in the volume of the medium; its contribution is constrained by the GWTC-3 tests of gravity (LIGO Scientific Collaboration et al., 2022).

The clearest measure in this picture is a comparison between two clocks. If they follow different trajectories and later reunite, the proper times they have accumulated may differ. The difference displayed on the clocks is an objective result of their histories through distinct gravitational environments. Locally measured rates, sizes, and frequencies remain relational readouts within the same medium.

2.5 Historical Context and the Place of RefG

The refractive reading of gravity has clear precursors in the history of physics. As early as 1911–1912, before general relativity was complete, Einstein related the coordinate speed of light to the Newtonian gravitational potential (Einstein, 1911, 1912). In 1851, Fizeau showed that moving water partially drags the propagation of light (Fizeau, 1851). Gordon later related the path of light through a dielectric medium to geodesics of an effective metric (Gordon, 1923). This established a bridge between the optics of media and gravitational geometry.

In 1957, Robert Dicke described gravity through a variable refractive index of the vacuum (Dicke, 1957). The modern program of analogue gravity began with William Unruh’s result that sound waves in a moving fluid follow an effective geometry analogous to a black-hole horizon (Unruh, 1981). Work by Volovik and by Barceló, Liberati, and Visser showed how such a geometry can emerge from the density and flow of a medium (Barceló et al., 2011). In 2019, a thermal analogue of Hawking radiation was also observed in a Bose–Einstein condensate (Muñoz de Nova et al., 2019).

Alongside medium-based models stands the thermodynamic approach to gravity. Sakharov interpreted gravity as an effect arising from quantum fluctuations of the vacuum (Sakharov, 1967). Jacobson showed that Einstein’s equation can be obtained from the thermodynamics of local horizons (Jacobson, 1995). RefG is related to this line of thought in treating spacetime geometry as emergent; in RefG, it is generated by a concrete pressure–refractive medium.

Refractive Gravity adds three fundamental elements to this rich historical line:

- **First—an ontological foundation:** The resonant self-distinction of one universal foundation produces a medium in which light, matter, measuring instruments, and the observer take form as physical modes. RefG is therefore not merely a mathematical analogy but an independent framework for physical research.
- **Second—a shared local resonant rhythm:** The collective cadence of oscillons turns the local unobservability expressed by relativity into an internal dynamical result of the system.

- **Third—a universal pressure-deficit channel:** RefG uses the pressure deficit as a common language of source and readout. In this language, mass, clock rate, light deflection, and the local tensor response are different readouts of one pressure–energy balance.

Within this historical map, the closest conceptual analogue to RefG is Erik Verlinde’s entropic gravity (Verlinde, 2011). In Verlinde’s model, Newton’s law arises from the entropic balance of information on a holographic screen. RefG shares the idea that gravity is emergent and takes the pressure deficit of the manifested medium as its source. The two mechanisms anticipate different observational outcomes and are tested independently.

3 Three Scales and Cosmological Horizons

3.1 The Local Scale: Compact Objects, Internal State, and External Readout

The pressure chain developed in the preceding chapters appears in nature on three scales. Locally, it is the deficit formed around an oscillon; galactically, it is an ordered flow of the medium; cosmologically, it is the long-term evolution of the common background. All three are distinct dynamical regimes of one foundation. We begin with the local scale, where the mechanism is most pronounced in compact objects.

In an extreme gravitational regime, an object’s *internal state* and its *externally read mass* become sharply distinct. A distant observer reads the object through its deficit profile, clock rate, and operational size. The interior geometry, the horizon, and the causal connections are determined by the complete solution.

In this two-channel picture, compression deepens the deficit gradient, slows the clock rate, changes the conditions under which light can escape, and reduces operational size. The internal state and the externally read mass are not the same quantity. During a merger, the local deficit deepens while outgoing energy reduces the remnant’s total mass as read from afar; their exact relation is fixed by the complete dynamical solution. Chapter II used GW150914 as an observational example of this balance.

The metric determines the difference between internal and external scales. On the compact branch of RefG, this difference may be large. The exact internal scale, the existence of a horizon, and the global causal structure are all determined by the complete solution.

From outside, such an object may appear small, dark, and almost frozen, even though the proper length of the same region is much greater. The local speed of light is invariant in both descriptions; only time and size as read from afar differ. The frozen state is the appearance perceived by a distant observer, while the complete causal structure decides whether a horizon exists. An intuition similar to a dark star appears as early as Michell’s work of 1784 (Michell, 1784).

RefG reads a singularity as a boundary of external readout: under a strong deficit, the externally read value of the internal scale tends toward zero, while the local size stops at a finite saturation limit and never shrinks to zero. A self-consistent integration of these two descriptions must keep the local density and curvature finite.

Penrose’s singularity theorem rests on specific energy and causal conditions (Penrose, 1965). On the compact branch of RefG, the effective source violates one of them—the null energy condition (NEC). The singularity conclusion therefore does not carry over directly to this branch. The existence of a regular interior regime must be established by a complete solution, correct boundary matching, and stability.

The concept of regular compact objects has also been developed in the modern literature. A well-known example is the gravastar of Mazur and Mottola—a condensate object without a singular core (Mazur and Mottola, 2004). Such alternatives may be distinguished through shadow size, mode spectrum, and echo signals (Cardoso and Pani, 2019). RefG is structurally

related to this family and connects its interior regime to the pressure–energy balance of the manifested medium.

RefG associates the feedback between mass and deficit with a finite equilibrium state. The field equations determine its existence and stability. In this picture, a compact object is a strongly loaded regime of the manifested medium whose internal state and mass inferred at large distance are registered through different channels.

3.2 The Galactic Scale: Vortical Flow and MOND

Observational law. Galaxy rotation curves do not decline as the distribution of visible baryonic matter alone would predict. At the same time, baryonic mass and rotation speed obey a tight relation—the baryonic Tully–Fisher relation. Historical landmarks are the rotation curve of Andromeda (Rubin and Ford, 1970) and the Tully–Fisher relation (Tully and Fisher, 1977). In the modern radial acceleration relation, 2,693 measurements from 153 galaxies fall on one common curve (McGaugh et al., 2016).

The standard Lambda cold dark matter model (Λ CDM) describes these phenomena through dark-matter halos. MOND instead identifies an acceleration threshold below which motion departs from the Newtonian expectation (Milgrom, 1983). Tensor–vector–scalar gravity (TeVeS) and its newer versions provide a relativistic account of this picture through additional vector and scalar fields (Bekenstein, 2004; Skordis and Złótnik, 2021). RefG associates the threshold with the natural scale of a large-scale coherent flow of the medium.

The vortical mechanism of RefG. The structure generating this map is a large-scale **vortical, or swirling, flow** of the manifested medium. An additional pressure deficit forms at the center of this ordered rotational regime, and baryonic matter gathers in the resulting well. The flattening of rotation curves and the Tully–Fisher relation are read as the combined refractive effect of a giant vortex in the medium and the imprints of baryonic oscillons.

The closest modern parallel to this picture is superfluid dark matter, in which a MOND-like force is associated with phonon excitations of a condensate phase (Berezhiani and Khoury, 2015). There, superfluidity is a phase of an additional particle component; in RefG, coherent flow is a regime of the unified manifested medium itself.

Morphology. A galaxy’s visible form may reflect the flow regime of the manifested medium. Ordered laminar rotation favors disk and spiral structures; a more chaotic or isotropic flow favors dispersion-supported systems. These are endpoints of a continuous spectrum, since some elliptical galaxies also retain substantial rotation.

Clusters. In galaxy clusters, modifying the acceleration law alone is often insufficient. The missing-mass problem was historically exposed in the Coma Cluster (Zwicky, 1933). RefG reads the cluster mass from three sources: the common deficit of the long-wavelength network; individual galaxies and their oscillon tails; and the cluster’s vortical flow together with the memory of past mergers. All three enter a single pressure-deficit balance, and each is counted only once. In Chapter IV, particles and the cosmic web will appear as the two ends of one spectrum.

Two discriminating observational signatures follow from this picture. A continuous field can also form a vortex, so swirling flow alone does not distinguish discreteness in the manifested medium. The signatures of a nodal microstructure are quantized circulation and a compatible dispersion relation.

First—quantization of circulation. If the flow of the manifested medium has a single-valued condensate phase, the expected unit of circulation is h/m , where m is the mass of the effective

flow mode. For an extremely small m , the imprints of individual vortices may appear in weak gravitational lensing or disk kinematics. Similar signatures are sought in the wave-like halos of ultralight dark matter (Hu et al., 2000; Hui et al., 2021).

Second—the frozen imprint of the Bullet Cluster. In the Bullet Cluster, the gravitational-lensing centers are displaced from the X-ray plasma at a significance of roughly eight standard deviations (Clowe et al., 2006). RefG uses this separation as the principal quantitative measure of a model of memory in the medium. A coupled model of the medium’s flow and the gas motion must determine the lifetime of that memory; the test data are the lensing map and the spatial distributions of gas and galaxies.

This mechanism has one unified observational test: the SPARC galaxy data, galaxy clusters, Bullet-Cluster-like mergers, the cosmic microwave background (CMB), and the large-scale structure of the universe (Lelli et al., 2016; McGaugh et al., 2016). If particle dark matter does not exist, the vortical source must also perform its cosmological role.

For the same metric, material content, recombination history, and initial conditions, RefG preserves the primary linear source terms of the CMB. This is a test of algebraic consistency. Without particle dark matter, its function must be taken over by the vortical and nodal channels. The final test is whether this replacement can recover the full statistical picture of the CMB and large-scale structure.

3.3 The Cosmological Scale: Relaxation of the Self-Distinguished Background and Two Languages of Time

At the cosmological scale, the ordinary metric description remains in force: the universe expands, distances between galaxies increase, and light undergoes redshift. RefG reads the same history from within the medium—as the gradual relaxation of the cosmological background. The long-range imprints of oscillons accumulate into a common background and alter its pressure. In the process description, earlier epochs correspond to a larger number of internal cycles available for the formation of structure.

Photons follow the light paths of the effective metric, while cosmological redshift is read through the standard ratio between the scales at emission and reception. In RefG language, this is a translation of phase: an imprint arriving from an earlier background is read against the local resonant rate at every epoch, while dimensionless ratios remain invariant.

The age of the universe—approximately 13.8 billion years—is metric time calibrated by present-day clocks (Planck Collaboration, 2020). Before the first stable material modes formed, no operational clock had been defined. Process time supplies an additional description of the phase ordering of self-distinguished regimes that were still pre-clock.

The idea of two time scales in cosmology has historical antecedents. Milne described one universe using kinematic and dynamical time (Milne, 1935, 1948). Wetterich related expansion to changing material standards (Wetterich, 2013). In RefG, process time describes the phase ordering of self-distinguished regimes and the accumulated number of internal cycles.

In the process description, each epoch measures the past using its own local clock. A future epoch may therefore read the duration of the past differently through its altered cadence. Metric age continues to increase in the ordinary way; the difference lies only in the rule used to count internal cycles.

The gradually fading vibration of a string makes these two languages vivid. An internal clock counts time by the number of oscillations, and its unit gradually lengthens; a fixed external clock records the same decay at a different rate. One physical history is read on two different measurement grids.

In early JWST data, massive red galaxies appeared approximately 500–700 million years after the Big Bang (Labbé et al., 2023). A substantial part of this tension is resolved by star-formation

efficiency, revised mass estimates, and the contribution of active galactic nuclei. The process budget concerns only the remaining moderate discrepancy.

Cosmological expansion is read in RefG at three descriptive levels:

1. **In the external metric language**, distances in space increase.
2. **In the material language**, clocks and atomic standards from different epochs occupy different local states of the self-distinguished medium.
3. **In the language of the medium**, the mean density of nodes and the connections among them change globally over cosmic history.

These are three levels of description of one history. The external metric is the primary language of observational calculations.

The oscillatory imprint locked to the source of every oscillon extends beyond its immediate environment and forms a long-range gravitational profile. The freely propagating share released during the formation, interaction, and reconfiguration of sources is distributed among many regimes and regions and becomes mixed in phase.

At the macroscopic level, reconstruction of this energy's initial configuration becomes progressively less likely. RefG associates the accumulated redistribution with relaxation of the mean pressure of the manifested medium and with increasing entropy: differences gradually diminish as the system approaches equilibrium. Complete equalization would erase the very difference that drives relaxation, so the process approaches this limit without ending at any finite time.

RefG associates the macroscopic arrow of time with the direction of relaxation of the cosmological background. The dissolving-crystal picture introduced in Chapter II makes this intuition vivid: color dispersed through water does not spontaneously gather back into a solid crystal. RefG reads this statistical irreversibility as the macroscopic direction of time.

In this picture, the thermodynamic, cosmological, and radiative arrows of time share a common source—background relaxation. At large scales, the process moves in one direction toward equilibrium (Planck, 1899). The arrow of time is thereby tied to the initial nonequilibrium state of the universe.

The late universe is expanding at an accelerated rate. Standard cosmology attributes this phenomenon to Λ or dark energy. RefG reads it as an energetic residue left by relaxation of the self-distinguished cosmological background. The stress of this residue produces negative pressure, while its slow variation yields an almost constant cosmological term. RefG relates the numerical value of Λ to the parameters of the relaxation equations.

Within a unified medium, the vacuum-energy problem takes a different form: the question is how the equilibrium energy of the medium enters the gravitational readout. In the conventional approach, the sum of zero-point energies of independent fields exceeds the observed value by roughly a factor of 10^{120} . In RefG, these channels belong from the outset to the common equilibrium of one medium, while the small cosmological residue is a deviation left from that equilibrium. Related mechanisms have been studied in condensed-matter physics and in q -theory (Volovik, 2003; Klinkhamer and Volovik, 2008).

The same medium therefore gives rise to both gravitational refraction and late-time cosmological relaxation. The model must be tested through the evolution of energy density and pressure, with baryon acoustic oscillations (BAO), Type Ia supernovae, and the CMB as observational anchors (Abdul Karim et al., 2025; Brout et al., 2022; Planck Collaboration, 2020).

Scope of process time. Process time is used for internal physical interpretation. The expansion history, the CMB, Big Bang nucleosynthesis (BBN), photon redshift, atomic clocks, and stellar nuclear processes are calculated in metric time. Supernova evolution also follows the

observed cosmological time dilation (Blondin et al., 2008). Primary observational calculations therefore remain in the metric language.

3.4 Quantum Randomness and Two-Level Time

These two levels of time also offer a new interpretation of quantum physics. The place and moment of a measurable event are registered in ordinary clock time. RefG associates outcome selection with a deeper pre-clock level. This level cannot be measured directly by a clock, because a clock is itself an oscillon structure of the manifested medium.

Violations of Bell inequalities rule out models in which outcomes are determined solely by preassigned local parameters (Bell, 1964; Hensen et al., 2015). At the same time, the no-signaling condition preserves causality: distant correlations cannot transmit controlled information faster than light.

RefG reads this as **pre-clock coordination**. At the deeper level, it is not a signal moving through space; in the language of clocks, it appears only as a coordinated relation between distant outcomes.

Wheeler’s delayed-choice experiment can be read in the same language (Wheeler, 1978). A late change in the measurement arrangement determines which result appears: interference in a closed interferometer, or path information in an open one (Jacques et al., 2007). RefG’s two-level time describes this without a signal sent into the past.

Several approaches in modern physics are related to this picture: the timeless fundamental level of the Wheeler–DeWitt equation; the Page–Wootters mechanism, in which time arises from correlations among subsystems; and Bohm’s pilot wave, which carries nonlocal coordination (DeWitt, 1967; Page and Wootters, 1983; Bohm, 1952). RefG differs through its concrete picture of one foundation and the medium that emerges from it.

This quantum picture has two strict quantitative criteria:

1. the **Born probability distribution**—the macroscopic statistical readout of internal coordination (Born, 1926);
2. the **no-signaling condition**—internal coordination cannot become a transmissible signal that violates relativistic causality.

4 Oscillons and the Origin of Matter

4.1 Many Patterns in One Ocean

For a vivid picture, imagine a single continuous ocean with waves rising within it. Most form, propagate, and soon disappear, but a few settle into stable, confined patterns: one may be ring-shaped, with the flow circulating continuously as a vortex; another may be a stable spherical pulsation; and a third, a spiral wave.

The ocean picture conveys the central proposition of this chapter: the electron, proton, photon, and quark are different resonant patterns of one universal manifested medium. The self-distinguished medium itself oscillates, and a particle is a stable form of that oscillation. A historical precursor is the Skyrme model, in which nucleons are read as topological solitons of a nonlinear meson field (Skyrme, 1961). RefG adds one step to this intuition: all such forms are modes of the same manifested medium.

Many distinct resonant modes can coexist within the medium generated by the self-distinction of one foundation. Each stable mode is read as a separate physical unit. Their physical multiplicity belongs to distinct states of one primary identity, while energy, momentum, and charge add within the balance of the complete system.

A particle is not a small solid sphere. The oscillatory amplitude of the medium extends through space, while the point registered in a detector is the local encounter between this wave



Figure 2: Resonant modes of the manifested medium. In the intuitive map of RefG, stable wave patterns in one medium are read as different particle modes: a spherical pulsation, a ring-shaped vortex, and a spiral wave. A particle is not a separate substance here; it is a locally confined resonant pattern of the manifested medium.

and the detector’s internal oscillon modes. RefG thereby associates the amplitude and phase of a quantum state with a real medium.

The electron cloud is the amplitude itself, not the hidden path of a small body moving rapidly around an orbit. The extended state carries a real phase structure through space, while the registration point arises through local resonant contact with the detector.

Single-electron interference joins two facts: the state’s amplitude depends on the geometry of both slits, yet each registration appears as one point on the screen. Tonomura’s group directly recorded the gradual build-up of this pattern (Tonomura et al., 1989). A stationary electronic state in an atom likewise does not radiate as a classical orbit would. RefG reads these results as the conjunction of an extended wave in the manifested medium and a localized registration.

The width of a wave packet is not the particle’s internal size. Scattering experiments measure the internal distribution of charge and energy; this is how, for example, the proton radius is determined. At high energies, the electron appears as a pointlike electromagnetic object down to scales of approximately 10^{-18} m (Navas et al., 2024). An oscillon model must produce a charge profile that reproduces this pointlike behavior throughout the measured range.

4.2 Localized Confinement: Massive and Massless Modes

In RefG’s particle map, the principal distinction between massive and massless modes is **localized confinement**.

The energy of a massive oscillon is confined within a localized form. When vortical circulation or stable pulsation closes a wave pattern within a finite region of space, it creates a pressure-deficit profile. Long-lived localized pulsations are known in nonlinear field theory as pulsions and oscillons (Bogolyubsky and Makhankov, 1976; Gleiser, 1994; Copeland et al., 1995). RefG associates their stability with population-wide resonant locking. In the external readout, this confined energy appears as inertia and gravitational mass.

Ordinary field-theoretic oscillons are long-lived but gradually decay through radiative leakage. In a stable RefG oscillon, mutual locking across the population suppresses this leakage while

preserving a stable oscillatory connection to the medium. The confined core and the pressure profile phase-locked to its source form one self-consistent configuration: the core preserves its identity, while its oscillatory imprint is read in the medium as ordinary gravity and does not require continuous energy loss. Strong wave emission appears when this equilibrium reconfigures rapidly.

A massless carrier mode has neither a rest frame of its own nor a stable localized core; it propagates through space at the limiting speed. RefG reads the photon as a propagating phase mode of this kind.

Helicity and rotation confined to one location are different. Photon helicity describes the helical phase structure along the direction of propagation and does not form a localized vortex. A massive mode has a stable core; a massless mode is a phase wave propagating through space.

A photon's energy and momentum enter the gravitational balance. Free radiation leaves no persistent localized deficit profile of the kind associated with a massive oscillon, but its energy contributes in full to the source of the complete system.

The same rule applies to composite particles. Strong-interaction energy confined within a hadron does not appear as a separate external imprint, but it contributes to the hadron's total energy and gravitational mass.

Form, spin, charge, color, and polarization distinguish particles from one another. Gravity reads their energy through one common pressure-deficit source. Binding energy, free radiation, and a particle's rest energy therefore all enter the gravitational source of the complete system.

4.3 Self-Forming Resonators and the Two-Level Filter

A wave equation permits an effectively infinite variety of forms, yet nature presents a sparse, well-defined spectrum of stable particles. Why does nature select this particular set from infinitely many possibilities? RefG answers through a **two-level filter**.

The first level is a filter of form and mutual compatibility. A handpan provides the intuition. In a conventional instrument such as a guitar, the resonant length of the string is fixed in advance; a three-dimensional oscillon, by contrast, must create its own resonant cavity within the medium. In a stable handpan note, the fundamental, octave, and upper fifth combine in the ratio one to two to three.

The oscillon's deficit changes the local geometry, and the altered geometry selects the possible modes in return. A mode must be able both to exist in the medium and to coexist with the rest of the particle spectrum. Stability arises from compatibility across the entire local system.

The second level tests energetic stability. If a form has an allowed decay path, it transforms into lower-energy modes. Hadronic resonances are familiar examples. The electron, as the lightest charged particle, is protected by charge conservation. The proton's long lifetime likewise requires a corresponding protection mechanism, which RefG's particle dynamics must explain.

Typical hadronic resonances live for roughly 10^{-22} – 10^{-24} seconds: they appear in the spectrum but cannot remain as long-lived members and pass into already allowed lower-energy modes. The energetic filter therefore does not erase the spectrum; it redistributes energy among stable channels.

The mathematical language of this filter is that of **attractors**—stable states toward which a system evolves naturally. Energetic conditions select which mutually compatible modes remain as long-lived particle states.

A stable particle is a self-consistent equilibrium: the wave form creates a deficit in the medium, and the deficit reinforces that same form through feedback. In nonlinear fields, Q-balls are protected by a conserved internal charge (Coleman, 1985). In RefG, additional protection arises from the common phase locking of the particle population. The gravitational feedback between mass and deficit belongs to the same equilibrium.

The self-distinguished medium is the common oscillatory channel, while phase locking occurs directly among oscillons. They synchronize with the existing spectrum through the fixed resonant

ratios described in Chapter I.

This mutual locking requires neither a master clock nor an external conductor. Huygens’s pendulums suspended from a common beam, phase-locked laser modes, and the Kuramoto model provide familiar examples of spontaneous synchronization (Huygens, 1665; Kuramoto, 1975). A historical relative is the bootstrap program in particle physics, in which the spectrum forms an ensemble of mutually compatible modes (Chew and Frautschi, 1961).

A slow change in the gravitational field preserves these locks. If the background changes much more slowly than a particle’s internal cycles, the mode adjusts smoothly to the new environment. A rapid and strong change may break the lock, and the released energy is redistributed among other compatible modes.

A wave entering a different pressure environment adjusts to the new local conditions. Its wavelength, process rate, operational size, confinement scale, and externally read mass all change together, while dimensionless ratios and phase harmony are preserved. Chapter I called this collective retuning of the particle configuration the transposition principle.

4.4 The Global Resonator: Unity of the Microscopic and Macroscopic Spectra

The two-level filter tells us *how* the particle spectrum is selected. We now ask: where does this selection occur? Ernst Chladni’s plate provides the intuition (Chladni, 1787). Sand scattered across a vibrating metal plate gathers along the nodal lines of a standing wave, where the motion is minimal. In the RefG map, this picture corresponds to global modes and their nodes. Every resonator has a spectrum of its own: geometry and boundaries determine which modes are amplified and which fade away.

RefG reads the measurable universe as a **global volumetric resonator**—a three-dimensional medium with its own allowed spectrum. Its geometry, nodes, and boundaries arise from relations within the self-distinguished medium and form a self-organized spatial order. A strong excitation in the early universe activates the spectrum as a whole. Some modes reinforce one another, while others cancel. Forms compatible with the geometry and background pressure become stable particles; the energy of the remaining forms passes into other allowed channels.

In this picture, today’s stable particles are the local, short-wavelength modes of the universal spectrum. Where phase coordination did not have time to complete in the early universe, frozen linear topological defects may have remained in the manifested medium. We return to this cosmological imprint in Chapter V.

Within the same map, the effective dispersion of the self-distinguished medium links frequency to spatial wavelength. Short modes correspond to localized particles; long modes correspond to the large-scale distribution of matter—the cosmic web.

Galaxies here resemble Chladni’s sand: their distribution reveals where the self-distinguished global medium has stable nodes, filaments, and vast voids. Galaxy clusters are the deepest nodes in this network, where long-wavelength resonance lowers the background pressure. This joins the picture of clusters developed in Chapter III.

The conclusion of this map is both simple and elegant: a microscopic particle and the macroscopic cosmic web are two readouts of one universal spectrum—a form bound locally on the inside and a pattern spread globally on the outside. In standard cosmology, the filaments, sheets, and voids of the cosmic web are described by Zel’dovich’s anisotropic collapse (Zel’dovich, 1970). RefG describes the same form in the language of a resonant medium.

RefG’s unified spectral map associates short waves with particles and long waves with the deficit of the cosmic web. This idea must account simultaneously for the particle spectrum, cosmic structure formation, cluster mass, and the BAO standard scale of roughly one hundred h^{-1} megaparsecs (Eisenstein et al., 2005). Only such a joint test can show whether these are truly two readouts of one spectrum or two independent descriptions.

RefG reads the measurable universe as a finite, boundaryless resonator. Wetterich’s model shows that a change of variables can recast expansion as a relation between geometry and

material standards (Wetterich, 2013). In RefG, the compatibility of these languages is tested through agreement among all dimensionless observables.

In the finite-resonator picture, its total size remains fixed while cosmological history is read through changes in internal standards. The CMB, Big Bang nucleosynthesis, and the redshift dependence of temperature are still calculated in the standard metric language. Any imprint of a finite cosmic size should appear at the largest cosmological scales and in global topology.

Present data show no clear discrete global spectrum. The anomalies found at the largest scales are weak, while searches for cosmological topology push the possible size of the resonator beyond the observable horizon (Planck Collaboration, 2016). Detection of a global cutoff would indicate finite size; failure to detect one only raises the lower bound on that size.

In the resonator’s internal language, the initial singularity is read as an extreme initial state of local standards. The question *What does the universe expand into?* no longer arises: no external vessel is required, because expansion is a relation between the resonator’s own geometry and its material measures.

4.5 Discreteness and the C_3 Symmetry of the Generations

In the Standard Model, the masses of the electron, muon, and tau are empirical parameters. RefG reads them as an internal, order-nine, three-phase (C_3) triplet of a single topological defect: three distinct resonant modes of one physical structure.

In RefG, the charged-lepton triplet is described by one common frequency scale and three values of a three-phase modulation. The three phases are separated by one third of a full cycle, while mass follows the square of the confined oscillation frequency. The depth of modulation and the initial phase determine the distribution of the triplet; common oscillon dynamics gives these parameters their physical meaning.

In RefG’s order-nine return map, the second step of the oriented charged branch fixes the phase at two ninths of a radian. When the modulation depth equals the square root of two, the three-phase algebra yields Koide’s empirical ratio of two thirds exactly (Koide, 1982, 1983).

Three-phase geometry is already known in the literature. Brannen described it through a circulant phase (Brannen, 2006), while Foot related Koide’s ratio to the 45-degree geometry of the square roots of the masses (Foot, 1994). In Brannen’s later model, two ninths of a radian remained a dynamical phase (Brannen, 2010). Shulga obtained the same magnitude in an independent cycle with the opposite orientation (Shulga, 2026). RefG associates this phase with the order-nine return cycle of one defect.

A simple three-phase-symmetric potential cannot select the initial phase. Its source must therefore be sought in a deeper topological cycle—a holonomy. The second modulation parameter describes an equal distribution of energy between the internal singlet and doublet channels. The phase and the modulation depth arise through different mechanisms.

Equalization of the internal channels fixes the modulation depth at the square root of two, while the second branch of the order-nine cycle fixes the phase at two ninths of a radian. Anchored by the measured electron mass, this branch misses the muon mass badly but lies close to the tau mass: the discrepancies are 451.7 and 0.61 standard deviations, respectively. This mismatch shows the size of the radiative correction required by the geometric map (Sumino, 2009).

The balance also has a simple musical intuition: the charged-lepton C_3 /Koide triplet resembles the cold, long-lived, mildly anharmonic resonance of a three-dimensional volumetric resonator. Its tones are different modes of one body.

5 Topology and Quantum Numbers

5.1 Phase, Orientation, and Topological Quantization

Until now, the manifested medium has been described chiefly in terms of amplitude and local pressure. Its complete state is richer: it has a phase, an oscillation strength, an internal form, and a spatial orientation. In this work, Φ is shorthand for that unified state, not merely a single, simple scalar quantity.

If the state is described only by a single-valued scalar and the flow is only its gradient, a vortex cannot arise. A topological vortex requires a multivalued phase, a line defect, and an internal orientation locked in space. The oscillation amplitude vanishes at the defect core, while the phase completes an integer number of full turns around it. Structures of this kind also occur in superfluids.

The cosmological relative of this structure is the **cosmic string**—a line defect frozen in during a phase transition in the early Universe. Its formation is described by the Kibble mechanism (Kibble, 1976). In the language of RefG, this is the “frozen seam” introduced in Chapter IV. The topology of the manifested medium determines whether such lines can form, while observations constrain their possible abundance.

Lord Kelvin’s work on vortex atoms and vortex motion (Thomson, 1867, 1869) led Peter Tait to catalogue aether knots and helped give rise to the mathematical theory of knots (Tait, 1877). RefG continues this historical line through pressure–energy balance and population-wide resonant locking.

This history also carries a clear lesson. Vortex atoms in an ideal fluid could explain neither a stable particle spectrum nor the atomic data. In RefG, the knot is only the structural skeleton of the form; its stability is determined by pressure–energy feedback and resonant compatibility with other modes.

Phase quantization appears when the field is continuous along a loop, the state returns single-valuedly after a complete circuit, and the loop belongs to a nontrivial topological class. Under these conditions, the phase can change only by an integer number of complete turns.

The Călugăreanu–White–Fuller relation applies to a closed, orientable, framed ribbon (Călugăreanu, 1961; White, 1969; Fuller, 1971; Ricca and Nipoti, 2011):

$$Lk = Wr + Tw \tag{1}$$

Here Lk is the total linking number, Wr describes the spatial writhe of the axis, and Tw its internal twist. Writhe and twist may be converted into one another, but their sum remains invariant under continuous deformation.

In hydrodynamics, helicity is a conserved quantity in an ideal fluid, and Moffatt related it to the linking number (Moffatt, 1969). Onsager and Feynman predicted the quantization of circulation in superfluid helium, and Vinen measured the quantum of circulation experimentally (Onsager, 1949; Feynman, 1955; Vinen, 1961). Integer phase winding is therefore part of direct laboratory physics.

In the RefG map, the principle “**topological locking is read as charge**” means that electric charge is the external readout of a locked phase–orientation path. The topological number specifies the discrete class, while the field dynamics describe its electromagnetic readout.

The Möbius strip does not lie within the direct domain of this relation; it is described through an orientable double cover. The spinorial intuition of that double cover will be used in the next subsection.

In the RefG map, spin and charge are two geometrical readouts of one double-cover structure. The half-turn determines the spinorial return, while the direction of the twist and the number of complete windings determine the sign and class of charge. Their geometrical source is shared, although the spin transformation and the magnitude of electric flux must be calculated separately.

This direction also has experimental analogues. Kelvin waves have been directly excited and measured on quantized vortices in superfluid helium (Minowa et al., 2025); similar axial waves have been observed in a classical water vortex (Barckicke et al., 2026). In both cases, the wave travels helically along the vortex axis, although its speed and frequency depend on the medium and the structure of the core. For RefG, the essential quantity to compare is the oscillon’s internal wave law: the radial mode is associated with mass pulsation, and the spiral mode with radiation.

5.2 Spin and Fermionic Statistics

A phase twisted through half a turn along the oscillon’s axis creates a double cover resembling a Möbius strip. One complete circuit reverses the orientation; two circuits restore the initial state. This is the defining signature of a spinor: a 360-degree rotation changes the sign of the state, whereas a 720-degree rotation restores it completely. RefG associates the electron’s half-integer spin with this geometry. Changing the topological class requires the amplitude to vanish at the core, the defect to reconnect, or the boundary condition to change.

This geometry also provides an intuition for fermionic exchange. When two identical double-cover modes exchange positions, their joint phase shifts by half a cycle. To recover the Pauli principle, that change must become a sign reversal of the two-particle wave function. Two identical fermions would then be unable to occupy the same state.

In a bosonic mode, the exchange phase does not change the sign of the state, so many bosons may occupy a single quantum state. A clear example is the phase-aligned, or coherent, photonic state of a laser.

This is the topological intuition behind fermionic exchange. In quantum field theory, spin and statistics are linked by a rigorous theorem based on Lorentz symmetry, positive energy, and local causality (Pauli, 1940). At the quantitative level, RefG must recover the same conditions and the antisymmetry of fermionic states.

5.3 Charge, the Photon, and the Electromagnetic Channel

RefG reads electric charge as the external imprint of an oscillon’s internal phase twist. The two opposite orientations of the twist correspond to positive and negative charge. Outside the oscillon, this imprint spreads in every direction, and its total flux through a closed surface determines the magnitude of the charge.

In this map, the positron is the topological mirror image of the electron. When a particle and its antiparticle annihilate, their opposite phase windings combine, and the released energy passes into the radiation channel. In a neutral mode, the contributions of opposite phase twists balance, and the net flux through a closed surface vanishes. RefG associates the neutrino with this neutral sector.

Turning the asymmetry angle by the same amount at every point leaves the physical picture unchanged. A measurable imprint appears when that angle varies in space. RefG reads the photon as a transverse, helicoidal wave of this variation: it propagates the phase change of the asymmetry and mediates the electromagnetic interaction.

The topology of the electromagnetic field also appears in classical Maxwell theory. The field lines of Rañada’s vacuum solutions form linked loops and are classified by the Hopf index (Rañada, 1989). RefG connects this mathematical picture to the medium-based channel of charge and electromagnetic dynamics.

The sign of the force is determined jointly by the mediator’s spin, the type of source, and the sign of the coupling. In the electromagnetic vector channel, like charges repel and opposite charges attract; in the tensor gravitational channel, positive energy produces universal attraction. In three spatial dimensions, the dilution of a spherical flux gives a $1/r^2$ law, while the corresponding potential follows a $1/r$ dependence.

Bjerknes hydrodynamic forces make the limit of this analogy particularly clear: bodies pulsating in phase in a fluid attract, whereas bodies pulsating in antiphase repel (Bjerknes, 1906). In electrostatics, the pattern of signs is reversed. Mechanical resemblance alone therefore cannot determine the sign of the force; that sign follows from the joint structure of the vector mediator, the source, and the interaction.

The fine-structure constant describes the dimensionless strength of the electromagnetic channel. In RefG, it specifies the relative response of the same node network to electric flux and to the transverse oscillation of the photon. This ratio remains unchanged under a common reconfiguration of the medium. A quantitative theory must recover Maxwell's and Gauss's laws, charge conservation, photon energy, bosonic statistics, the two helicities, and their mirror relation.

5.4 Color, Confinement, and Three Generations

Three-dimensionality plays a structural role in RefG's particle map. The three spatial directions provide a geometrical carrier for the internal color triplet. Color remains an internal degree of freedom; the spatial axes form its geometrical basis.

A structure oscillating along a single axis is topologically incomplete and has open ends. Combining all three axes corresponds to baryonic closure, while joining an axis to an anti-axis corresponds to mesonic closure. An isolated quark cannot form a closed state; RefG associates color confinement with this geometry. Quarks carry three color states, gluons mediate transitions between them, and leptonic modes are color-neutral.

Here D_3 denotes the discrete deformation used in the eight-mode count. The resulting eight modes numerically match the eight generators of $SU(3)$ and form a structural map of the color sector. The criteria that distinguish a full $SU(3)$ structure are the algebra of these modes, local symmetry, and their couplings.

The question of three generations is the natural continuation of the C_3 triplet discussed in the preceding chapter. In the RefG reading, charged leptons are the ninth-order internal three-phase modes of a single topological defect, while the quark generations carry the same architecture into the colored axial sector. At this level, a generation is not a repetition appended externally to the particle table; it is a recurring mode of internal topological locking in the manifested medium.

A familiar example of holonomy is Berry's geometrical phase: after a complete cycle, a state acquires an additional phase determined by the geometry of the path it has traversed (Berry, 1984). RefG connects this idea to the return cycle of a single topological defect. Berry's result supplies the type of mechanism; the specific phase of two ninths of a radian is selected by RefG's ninth-order map.

Color and generation are distinct structures. Color indicates which internal axial direction is active; generation indicates which stable mode of the same resonator has been selected. Their triplicity is a numerical correspondence, not a structural identity.

In RefG, three-dimensionality is linked to the stability of knotted loops. One-dimensional closed loops form nontrivial knots precisely in three spatial dimensions; in higher dimensions, additional directions become available through which they may be untied. This argument singles out three-dimensionality in models where matter is built from such loops (Ehrenfest, 1918; Whitrow, 1955; Berera et al., 2017).

The same stability filter therefore makes two selections: it chooses the particle spectrum from mutually compatible closed forms, and it singles out three spatial dimensions as the setting in which those forms receive topological protection. In RefG's synthetic picture, the discreteness of matter and the three-dimensionality of space are two consequences of one locking principle.

5.5 A Geometrical Map of the Standard Model

5.5.1 The Electron and Leptons

In this map, the electron is a toroidal structure with a Möbius-type twist, in which all three spatial axes close. RefG associates this double cover with the Dirac gyromagnetic g -factor. The measured factor is slightly greater than two; Schwinger calculated the first quantum correction to this small difference, while modern experiments measure it to approximately the thirteenth decimal place (Schwinger, 1948; Fan et al., 2023). This number is one of the most stringent quantitative targets for the topological map.

5.5.2 Quarks and Color Charge

In this map, quarks are axial segments with open ends. Because they cannot form a closed structure on their own, they do not occur in isolation—the geometrical counterpart of color confinement. RefG associates fractional charge with dividing a complete winding into three parts and locking the topological phase in thirds. Color and electric charge are distinct channels, so the fractional charge of a quark requires an independent normalization. Color is an internal axial asymmetry, while the gluon carries transitions between color states.

5.5.3 Photons and Bosons

The photon is a massless, helicoidal transverse wave without localized confinement. RefG associates its two helicities and $U(1)$ structure with the transverse mode of the manifested medium. The eight-mode map of the gluons was described in the preceding subsection. In the RefG map, the W and Z bosons are locally confined, massive mediator modes.

RefG describes beta decay as a topological rearrangement: the confined structure changes the distribution of winding, couples to the charged weak channel, and releases a neutral, incompletely confined mode. The physical content of this picture brings the chirality of the weak interaction, energy–momentum balance, and the decay spectrum into a single geometrical process.

5.5.4 Neutrinos and the Higgs Boson

In this map, the neutrino is an electrically neutral, incompletely confined mode. Incomplete confinement is associated with its small mass and weak interaction, while flavor oscillation is associated with phase mixing among internal modes. The Higgs boson is read as a nonrotating spherical pulsation, corresponding to spin zero. Quantitatively, this picture must recover neutrino masses and mixing, the Higgs mass of 125 GeV, the vacuum expectation value, and the Yukawa coupling constants.

5.5.5 The Graviton and the Weinberg–Witten Theorem

RefG’s local tensor branch has the two polarizations of a gravitational wave. The Weinberg–Witten theorem places severe constraints on composite, massless, high-spin states in a Lorentz-covariant theory (Weinberg and Witten, 1980). RefG must therefore examine each assumption of the theorem separately. The essential features of the low-energy limit are a massless tensor wave, universal coupling, and effective Lorentz symmetry.

In this picture, the graviton is not a separate fundamental substance, but the quantum of a collective tensor displacement of the manifested medium. A simple acoustic scalar oscillation is insufficient: a gravitational wave requires tensor polarizations and universal coupling to every form of energy. RefG does not postulate the foundational level as a Lorentz-covariant field from the outset; it must instead recover Lorentz symmetry fully in the low-energy oscillon regime.

5.5.6 Dynamics and the Lagrangian

The mathematical language of this map is the Lagrangian—the dynamical law from which the field equations follow and the admissible forms are determined. Even a simple local field model can admit stable knotted solitons (Faddeev and Niemi, 1997). RefG connects this mechanism to the particle spectrum, masses, and decay rules.

5.6 Fundamental Constants: α and the Gravitational Hierarchy

The fine-structure constant, $\alpha \approx 1/137$, is one of the best-known mysteries in physics—Feynman regarded it as one of the great unexplained numbers of theoretical physics (Feynman, 1985). In the RefG picture, α is a dimensionless, invariant ratio between the electric and magnetic responses of the node network of the manifested medium.

The origin of the fine-structure constant must follow from an independent physical mechanism. Simple ratios of Planck-scale and electron-scale masses do not yield 137. The familiar ratio of the Compton wavelength to the classical electron radius reproduces this number only because of how those quantities are defined. Integer-valued steps in phase locking are genuine physics, as the Josephson effect and Shapiro steps demonstrate (Josephson, 1962; Shapiro, 1963). RefG therefore seeks the fine-structure constant not in an absolute frequency, but in a dimensionless internal ratio of the network.

The node network carries electric flux and the photon’s transverse oscillation as two responses of the same microscopic dynamics. The fine-structure constant measures their relative strength. When the entire network strengthens or weakens, both responses change together and their ratio is preserved. The constant is therefore not an absolute frequency or energy, but an internal proportion of the network.

The second fundamental question is the **hierarchy problem**: why is gravity so weak? Between two protons, the electromagnetic force is approximately 10^{36} times stronger than gravity; for a pair of electrons, the ratio is of order 10^{42} . The connection between these large numbers became the historical point of departure for Dirac’s hypothesis (Dirac, 1937).

RefG associates the hierarchy with the structural difference between two channels: **direct phase–transverse interaction** and a **secondary pressure imprint**.

- The **Planck mass** is the natural reference scale for gravitational interaction.
- The **electron and proton** are microscopic oscillons whose masses lie many orders of magnitude below the Planck mass.

The gravitational channel is the long-range pressure imprint produced by an oscillon’s energy, whereas the electromagnetic channel is a direct phase–transverse interaction.

For two electrons, the electromagnetic force is overwhelmingly stronger than gravity. This difference is governed by the fine-structure constant and by the enormous ratio of the Planck mass to the electron mass; in the force ratio, that mass difference is squared. RefG seeks the basis of this hierarchy in the common origin of gravitational interaction, the Planck scale, and oscillon energy.

In this map, the weakness of gravity reflects the distinct physical mechanisms of the two interaction channels. A quantitative account must bring particle masses, the photon mode, and charge normalization into a single system.

6 Epilogue: Limits of Knowledge

6.1 Why One Foundation and One Manifested Carrier?

Let us return, in closing, to the central question: why one foundation and one manifested carrier? A common carrier reduces the number of initial assumptions, but simplicity alone

cannot establish a physical theory.

The deeper issue is the **origin of regularity**. The electron and the photon are different resonant modes of the same manifested carrier. In RefG, the conservation of energy, momentum, and charge follows from their common dynamical balance: an interaction is a coupling between modes, while a conservation law is the geometrical expression of that balance.

Eugene Wigner wrote of the “unreasonable effectiveness” of mathematics in the natural sciences (Wigner, 1960). A common carrier gives this order a physical basis: shared mathematical laws reflect a shared dynamical structure.

Invoking a common symmetry among many independent fields merely moves the question back one step: where does the rule that unifies those different fields come from? In the picture of one foundation and the medium generated from it, that rule is the dynamics of the common carrier.

The argument also has an observational dimension. The electrical neutrality of macroscopic matter, the stringent upper bound on the photon mass, and the pattern of fractional quark charges call for a common structural explanation (Navas et al., 2024). RefG associates this order with the geometry of different modes of a single carrier.

The self-distinguished manifestation of one foundation gives rise to many physical properties: pressure, shear, phase, topology, rotation, memory, and resonance. This echoes Philip Anderson’s dictum, *More is different*: each new level of organization has laws of its own (Anderson, 1972). In RefG, these levels are connected by the dynamics of one multichannel medium.

RefG describes the measurable cosmological Universe as a finite but boundaryless global resonator. The discrete spectrum discussed in Chapter IV links microscopic and macroscopic scales. Its physical imprint should appear in cosmological modes and in the observable topology of the Universe.

The foundation of RefG is pre-spatial and therefore has no spatial size. It is the global resonator generated from that foundation that is finite and boundaryless. Its physical meaning is expressed by its own spectrum and by the imprint left within the horizon: the Universe measures itself through its own resonances.

6.2 Three Limits of Measurement

The Universe is always measured through its own constituents: our clocks, rulers, detectors, and bodies are all oscillon modes of the manifested medium. This view of a “self-measuring Universe” is related to Wheeler’s *It from Bit* program (Wheeler, 1990). The difference is that information is primary for Wheeler, whereas the foundation is primary in RefG; the informational imprint appears only in the self-distinguished medium.

Here Gödel’s theorem serves only as a logical analogy. In any sufficiently strong and consistent formal system, there are propositions that cannot be decided from within the system (Gödel, 1931). A corresponding question arises in a self-measuring Universe: how completely can one part of the system measure its own foundation and the limits of the whole? RefG distinguishes three such limits:

Limit of time: Before the first resonant clocks, measurable duration was undefined. “13.8 billion years” is metric time measured by present-day clocks. The foundation itself is not assigned an age in clock units, because duration is defined only through stable material cycles.

Limit of space: The foundation is not an object situated within measurable geometry and therefore has no spatial size. We measure only the relations between distinguishable nodes of the manifested medium. The cosmological horizon is the measurable boundary of this network; what lies beyond it is not directly observable.

An indirect imprint may nevertheless exist. If the global resonator is comparable in size to the horizon, its longest allowed mode will appear in the largest structures of the cosmological background and in the topology of the Universe. That imprint must be measured within our horizon.

Limit of scale: In RefG, the Planck length is the natural transition scale defined by c , $\hbar$, and G . Quantum and gravitational effects become important together in this regime. The “smallest” known scale has changed repeatedly in the history of physics—from the atom to the nucleon, and then to the quark. A deeper dynamical layer may exist below the Planck scale, or no new structure may appear there at all. Larger organized structures may likewise exist above our scale. Both possibilities must be judged solely by measurable traces.

In all three cases, infinity may arise from the relation between the rule of measurement and the demand for a complete description. Even a finite object can unfold without bound in a chosen representation: stereographic projection maps a finite sphere onto an infinite plane. Mandelbrot’s “coastline paradox” makes the same point in another way, as the measured length increases when the measuring scale is reduced (Mandelbrot, 1982). These examples remind us that the resulting picture depends in part on the form of description and the scale of measurement.

6.3 Methodological Principle

The methodological principle of this article is clear: an intuitive physical picture must become a rigorous, calculable, and falsifiable formalism. The analysis of a particle oscillon follows this sequence:

The text clearly distinguishes a working hypothesis from a computational result: a result is always tied to a specific code, the input data used, and the output obtained.

1. **Experimental basis**—record the mass, charge, spin, lifetime, and other observables.
2. **Mode identification**—select the corresponding resonant class of the manifested medium.
3. **Initial geometrical form (ansatz)**—describe the oscillon’s form, topology, and boundary conditions precisely.
4. **Dynamical law**—derive the action, field equations, and source.
5. **Symmetries and conservation laws**—determine the charge, spin, color, and zero modes.
6. **Existence and stability**—test the finite-energy solution, matching, and perturbation spectrum.
7. **Parameter closure**—determine every coefficient independently of the data we seek to predict.
8. **New prediction**—calculate an independent observable and its uncertainty.
9. **Data test**—compare the result with observations using a predefined falsification criterion.

The aim of this sequence is to derive the gyromagnetic g -factor from oscillon dynamics and to turn the reading of topological locking as charge into an independently testable law.

The same procedure must test the geometrical origin of hypercharge and isospin, the mechanism of color confinement, the evolution of the background temperature, and the universal coupling of every material mode to a single effective metric. Possible traces of a preferred frame and spatial anisotropy must be tested separately. If the intuition cannot be converted into a calculable form, it must be revised.

6.4 Falsification Criteria and Failure Modes

The list of working tasks states what we must do; the falsification map states what nature can remove. The outcomes divide into three levels: the entire framework may be rejected, a particular physical sector may fail, or only a specific geometrical reading may fail. This distinction matters, because the failure of one branch does not automatically nullify every other idea, whereas a

violation of the common foundational principle cannot be repaired by reworking an individual branch.

The entire framework is rejected if any of the following conditions is met:

1. Effective Lorentz symmetry and the isotropy of the speed of light fail to emerge in the oscillon sector. Optical resonators constrain anisotropy at the 10^{-17} level (Herrmann et al., 2009), while the gravitational and electromagnetic signals from GW170817 tie the tensor-wave speed to the speed of light at approximately the 10^{-15} level (Abbott et al., 2017).
2. The inertial and gravitational responses tied to the same total energy leave a composition-dependent residual. The MICROSCOPE experiment constrains such a difference at the 10^{-15} level (Touboul et al., 2022).
3. Process time must be added to the expansion history, the CMB, BBN, photon redshift, or the $(1+z)$ time dilation of supernovae. The process language cannot replace the metric calculation of these channels (Blondin et al., 2008).
4. Dimensionless ratios change under a common transposition. Quasar spectra test $\Delta\alpha/\alpha$ at the part-per-million level (Kotuš et al., 2017); such a change rejects the principle of the local collective rhythm.
5. A self-consistent bridge from one undifferentiated foundation to mutually distinguishable modes, local causality, and the additive accounting of energy, momentum, and charge cannot be derived. Without this bridge, the ontological core cannot close as a physical theory.

A particular sector is rejected if:

1. The source–flux normalization of gravitational waves fails to recover the quadrupolar energy loss tested by binary pulsars to a precision of 10^{-3} – 10^{-4} , or a scalar breathing-mode polarization exceeds the observational bound (Kramer et al., 2021; LIGO Scientific Collaboration et al., 2022).
2. The galactic-vortex map fails to describe the SPARC data and the radial acceleration relation (RAR), the mass budget of clusters, and the approximately 8σ separation between the lensing centers and the X-ray plasma in the Bullet Cluster (Clowe et al., 2006). Such a result rules out the vortex sector proposed as a replacement for dark matter, not the refractive weak-field chain.
3. With the same metric, matter content, recombination, and initial conditions, the primary CMB source terms are not preserved, or the combined vortical–nodal sources proposed as a replacement for dark matter fail to recover the statistical picture of the CMB and large-scale structure (LSS), together with the BAO standard ruler (Eisenstein et al., 2005). The first result rejects the consistency bridge; the second rejects the cosmological replacement without dark matter.
4. A specific compact-object branch fails to match the shadow size, mode spectrum, and bounds on echo signals. Such a result rejects precisely the internal geometry whose external phenomenology was tested (Event Horizon Telescope Collaboration, 2019; Cardoso and Pani, 2019).

A particular geometrical reading is rejected if:

1. An extended oscillon profile of the electron fails to recover the observed pointlike form factor.

2. The Koide/ C_3 map fails to find a definition of mass and a radiative bridge that bring the exact branch $(A_K, \theta) = (\sqrt{2}, 2/9)$ into agreement with the measured lepton masses. A fourth sequential charged lepton rejects the strict, exhaustive three-mode version of this map.
3. Once the effective mass and the coherence window have been fixed independently, circulation in the galactic flow is not quantized in units of h/m . This result rejects the particular reading in terms of a discrete vortical medium, not the existence of a continuous vortex.
4. The dispersion of Kelvin waves derived from the oscillon action fails a specified comparison with classical and quantum vortices.
5. The Möbius geometry fails to produce the Dirac value $g = 2$, or its quantum completion fails to recover the radiative corrections of the electron. At the same level, failure to recover the antisymmetry of Fock space nullifies the geometrical mechanism for the Pauli principle.

Each item has a specific measurement or a decisive calculation. This is why the list marks the boundary of the research program: a negative answer from nature removes the corresponding idea from the program rather than merely changing its wording.

Conclusion

This work has developed a unified conceptual chain for Refractive Gravity (RefG). The picture is as follows. There is one primary identity, prior to space and prior to clocks. Its resonant self-distinction gives rise to the measurable beginnings of multiplicity, space, time, size, and mass. Within the self-distinguished medium, a locally confined resonant form—the **oscillon**—propagates its oscillatory imprint throughout the connected medium and creates a pressure deficit. The stable profile of this same mechanism, phase-locked to its source, is ordinary gravity; the powerful, coherent component released during a rapid reconfiguration is a gravitational wave. An external observer reads the deficit profile as effective mass. Motion within the effective geometry altered by that deficit is gravitational attraction—the universal refractive response to the state of the medium.

This unified pressure chain unfolds across three principal scales:

- at the **local level**—in compact objects, where internal state and external readout are clearly separated;
- at the **galactic level**—in the large-scale vortical flow of the manifested medium, which RefG connects to MOND phenomenology;
- at the **cosmological level**—in the historical, asymptotic relaxation of the self-distinguished background.

The allowed spectrum of the manifested medium and the mutual locking of its modes connect the small, discrete set of particles to a single mechanism. The topology of phase and orientation gives quantum numbers a geometrical language. In the same picture, the fine-structure constant α is an invariant ratio between the electric and magnetic responses of the node network, linking the oscillon map to electromagnetic interaction.

The foundation of this work is the weak-field regime, where gravity has been tested most precisely. On the selected branch, RefG’s refractive description agrees at first post-Newtonian (1PN) order with the standard result of general relativity. From there, the investigation proceeds to particle, galactic, and cosmological scales. At every stage, the physical explanation is tied to a completed calculation and a predefined observational test.

The value of this picture lies not only in its unity. Every important connection is subject to a specific test: Lorentz symmetry, equivalence, wave-energy balance, galactic dynamics, the particle form factor, and topological numbers. Measurement ultimately assembles the picture; whatever fails to match nature's answer must be changed.

The aim of Refractive Gravity is to derive from one foundation a unified, calculable, and observationally testable account of gravity, matter, and measurability. This text establishes the basis of that research program and defines the path ahead.

Current Status of the Research Program

Current status information is gathered here in one place so that the main text can preserve the continuity of the physical narrative.

1. **The ontological foundation and the origin of measurability**—the ontological hypothesis of one primary identity, prior to space and prior to clocks, has been formulated. Its resonant self-distinction gives rise to distinguishable nodes, quantity, operational spacetime, size, material multiplicity, and local causality. A complete dynamical derivation of this transition is the next stage of the research program.
2. **Inter-node separation and minimum distinguishability**—the idea of operational distinguishability has been formulated; the carrier's microdynamics, the continuity law of the deficit, and the minimum distinguishability scale ℓ_* remain to be derived.
3. **The local collective rhythm**—common-transposition invariance and trial models of mutual locking have been obtained. The next stage is to formulate the microscopic law of inter-node transmission, establish the quantitative relation between density and coupling, and calculate the stable oscillon spectrum.
4. **Oscillon deficit, size, time, and mass**—the metric filters and the first-order biconformal branch have been obtained symbolically; a finite-energy nonlinear oscillon profile and the complete mass-readout map remain open.
5. **Weak-field gravity**—on the unloaded branch of the pressure-deficit channel, $H = 0$, $\gamma = 1$, and $\beta = 1$, while the Einstein and medium-field equations close consistently through quadratic weak-field order. An independent rederivation confirms the existence of this static exterior branch. It has not yet been shown, however, to be the unique solution selected by a finite source, and the static spherically symmetric case determines only γ and β among the ten standard PPN parameters. Finite-source matching, a complete analysis of physical degrees of freedom and stability, preferred-frame effects, the full inhomogeneous PPN derivation, transformation to standard gauge, and a likelihood-based test against data remain open.
6. **Inertia and gravitational mass**—the Noether identity has been obtained conditionally for a finite-energy, Lorentz-covariant localized solution satisfying the Laue boundary condition; a direct derivation of the zero mode remains open.
7. **Gravitational waves and local unobservability**—in the local Minkowski limit, the tensor propagation speed and the transverse-traceless detector response have been obtained; cosmological propagation, source-flux normalization, the complete waveform, and scalar polarization remain open.
8. **Compact objects**—a static, conditional map and the finite-core matching of the C^2 ansatz have been obtained; in one specific set of exterior-layer and pressure-reversal criteria, the admissible domains do not overlap. The physical equation of state, collapse, rotation, and full stability remain open.

9. **The galactic vortex and MOND/Tully–Fisher**—the transition algebra and the baryonic Tully–Fisher relation (BTFR) follow conditionally after the assumption $a_0 = cH/(2\pi)$ and vortex closure; derivation from the action, SPARC/RAR, the coefficients of the three source channels in clusters, and complete modeling of clusters remain open.
10. **Cosmological relaxation and dark energy**—the Friedmann–Lemaître–Robertson–Walker (FLRW) stress algebra and a conditional Λ -like diagnostic have been obtained; pressure relaxation of the manifested medium is an interpretive bridge. Perturbations and numerical fits to BBN, CMB, Planck, and BAO remain open.
11. **Two-level time and quantum randomness**—the interpretive map is compatible with the facts established by Bell tests and delayed-choice experiments; the Born probability distribution, the no-signaling condition, and the complete dynamics of measurement have not been derived.
12. **Particles as oscillon modes**—the geometrical placement map and initial spectral scheme have been formulated as candidates; a finite-energy oscillon, its form factor, masses, couplings, and decays remain open.
13. **The Koide/ $C_3/2/9$ structure**—the C_3 algebra gives $K = 2/3$ exactly; with $(A_K, \theta) = (\sqrt{2}, 2/9)$ and the electron pole mass as the anchor, the result misses the muon mass by 451.7σ and the tau mass by 0.61σ . The choice $h = 2$, the relation $m \propto \nu^2$, derivation from the action, and radiative protection remain open.
14. **Spin, Pauli exclusion, and fermionic statistics**—the Möbius double-cover geometry provides a candidate mechanism; a complete derivation of Fock space and of the spin–statistics relation of quantum field theory remains open.
15. **Charge, the photon, and the electromagnetic channel**—subtheorems establishing local-frame $U(1)$ symmetry, two helicities, a screened far field, and a conserved phase current have been obtained; canonical charge normalization, Maxwell’s equations, Gauss’s law, the Ward identities, quantum electrodynamics (QED), the Dirac sector, and the exact pressure-deficit channel through which free radiation sources gravity remain open.
16. **Color, quarks, and gluons**—the three-color, eight-mode construction provides only a partially compatible map; the structure constants of $SU(3)$, interaction vertices, scale evolution, the confinement potential, and fractional charge remain open.
17. **The electroweak map and the Higgs**—this is a geometrical placement map; $SU(2)_L$ chirality, a W/Z mass matrix that preserves a massless photon, CKM/PMNS mixing, the neutrino mass ordering, Yukawa couplings, and the vacuum expectation value have not been derived.
18. **The fine-structure constant α** —a node-based interpretation and a target-independent spectral measure have been obtained: α is read from the relative strengths of the electric-flux and photon responses, while the overall scale cancels. A compact microscopic action for the physical charge Q , a unique numerical ratio, the complete charged spectrum, and evolution to the Thomson limit remain open. The earlier boundary formula remains a target-dependent conditional result.
19. **The gravitational hierarchy**—the force ratio can be written algebraically as $F_{\text{em}}/F_g = \alpha(M_P/m)^2$; the origins of G , M_P , the oscillon’s microscopic energy scale, and the mass hierarchy remain open.
20. **A finite, self-measuring manifested Universe**—the global resonator is a hypothesis at the manifested cosmological level; low-multipole and topological signatures are candidates,

and no unique likelihood agreement with the data has been obtained.

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